



Machine Learning

Logistic Regression Classifier

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https://github.com/safayani/machine_learning_course

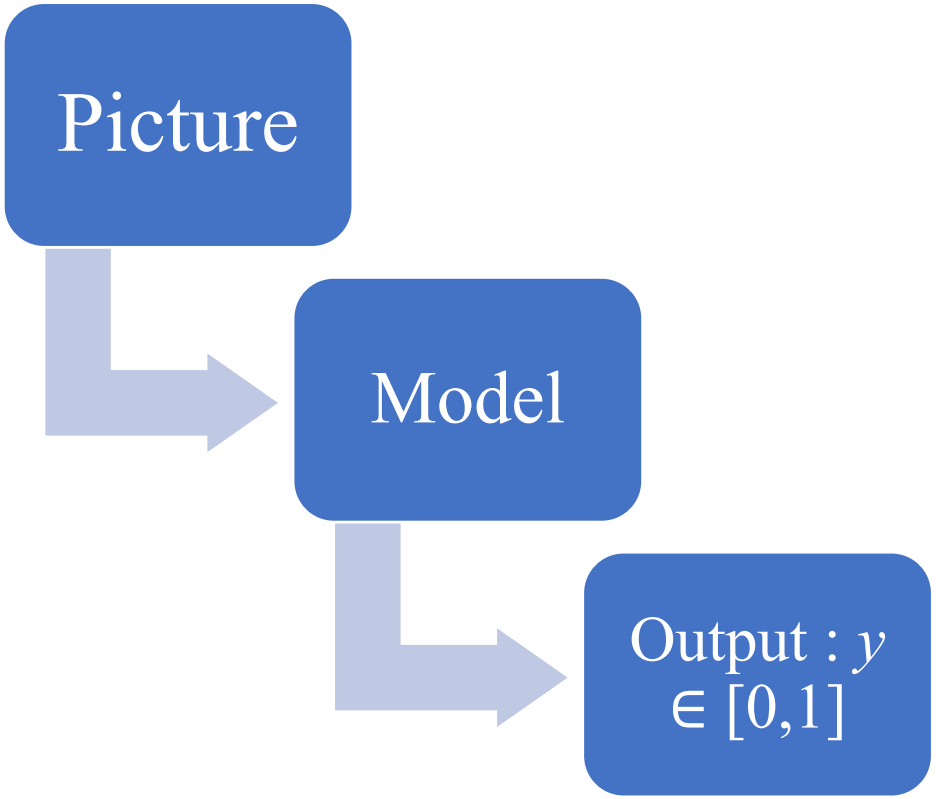
https://git.iut.ac.ir/safayani/courses/-/tree/main/Machine_Learning_course

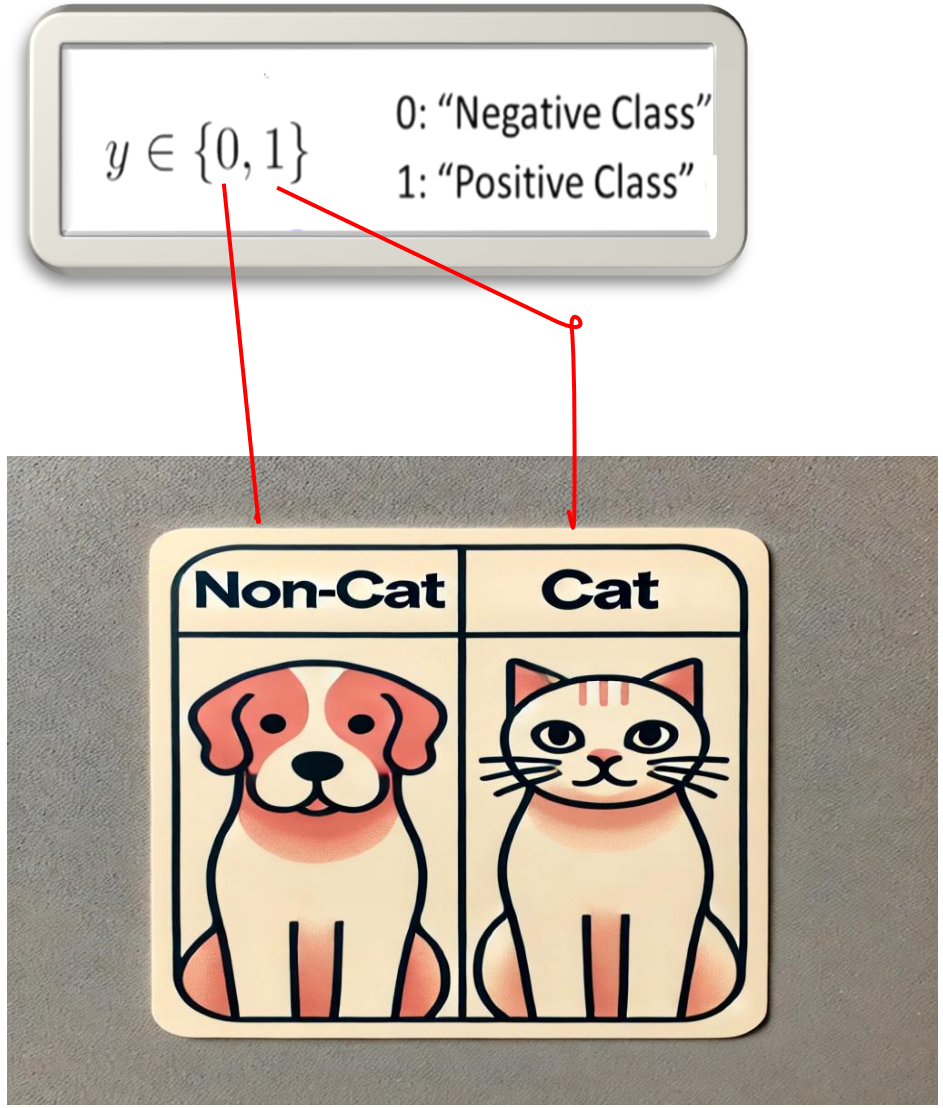


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Classification

- Email: spam/not spam
- Animal: cat/non cat





Logistic Regression

مبانی رگرسیون لجستیک:

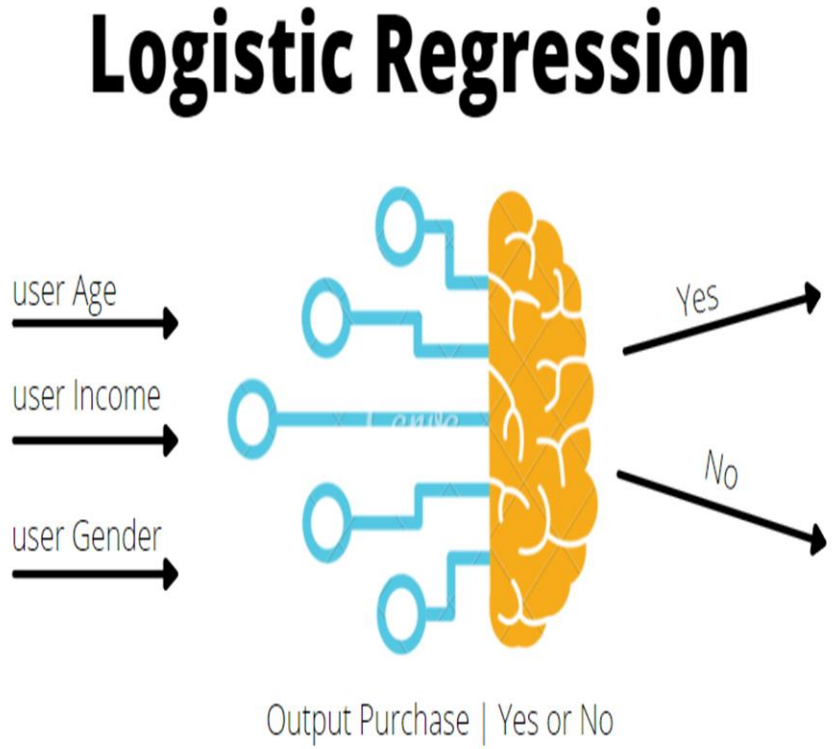
- عمدتاً برای مسائل طبقه‌بندی دودویی استفاده می‌شود.
- احتمال تعلق یک ورودی به یکی از دو کلاس را مدل‌سازی می‌کند.

مرزهای تصمیم‌گیری خطی:

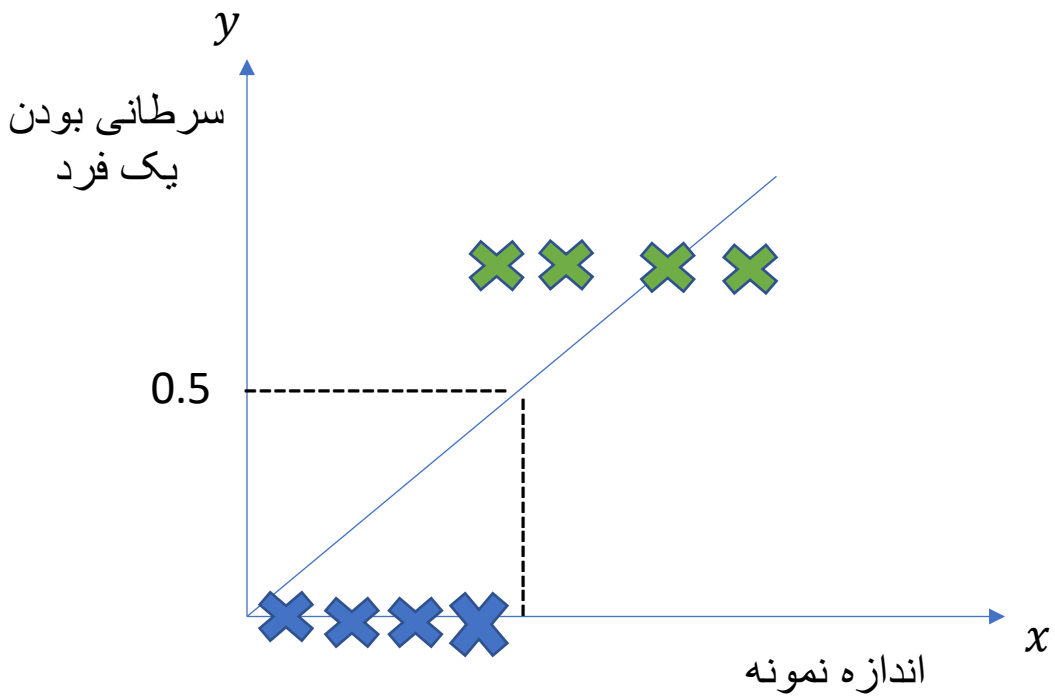
- رگرسیون لجستیک به‌طور معمول مرزهای تصمیم‌گیری خطی را در فرم استاندارد خود ایجاد می‌کند.

کاربردها:

- در مسائلی مانند طبقه‌بندی تصویر، تشخیص پزشکی و همچنین جایی که سطوح تصمیم پیچیده برای بهبود دقت ضروری هستند، استفاده می شود.



Classification



$$If: h_\theta(x) \geq 0.5 \rightarrow predict, y = 1$$
$$If: h_\theta(x) \leq 0.5 \rightarrow predict, y = 0$$

$$0 \leq h_\theta(x) \leq 1$$
$$0 \leq h_\theta(x) = P(y = 1)|x) \leq 1$$

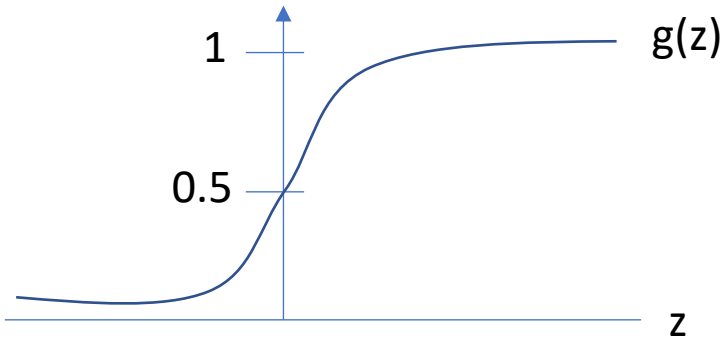
Sigmoid Function: Logistic Function

Want $0 \leq h_\theta(x) \leq 1$

$h_\theta(x) = g(\theta^T x)$

$g(z) = \frac{1}{1 + e^{-z}}$

$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$

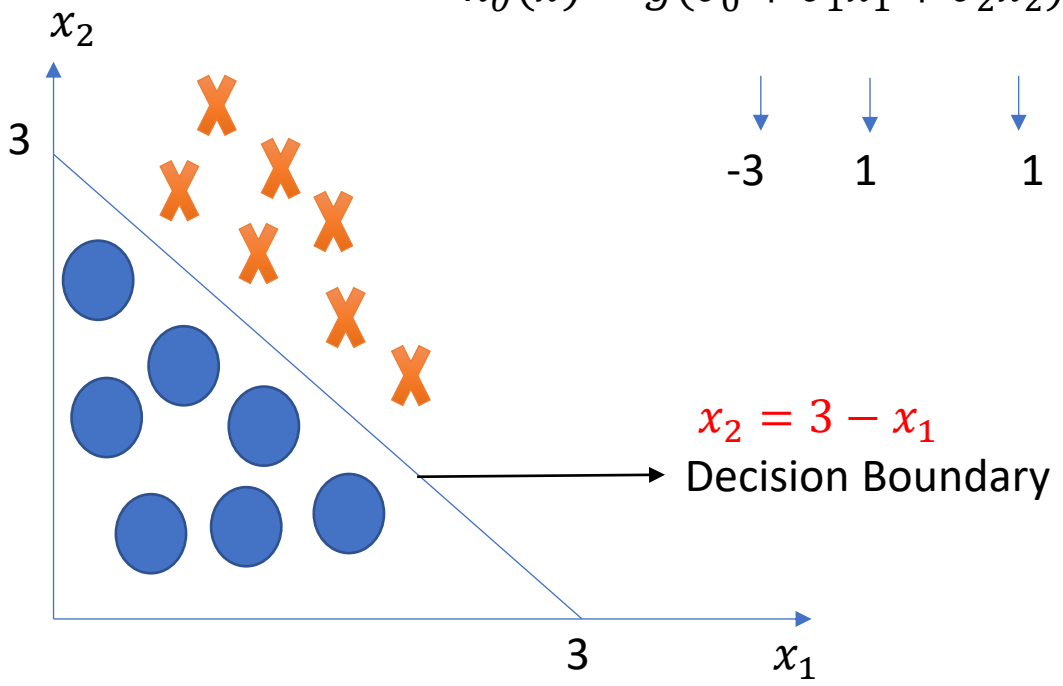


Model output example : $P(y = 1|x) = 0.8$

Decision Boundary

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

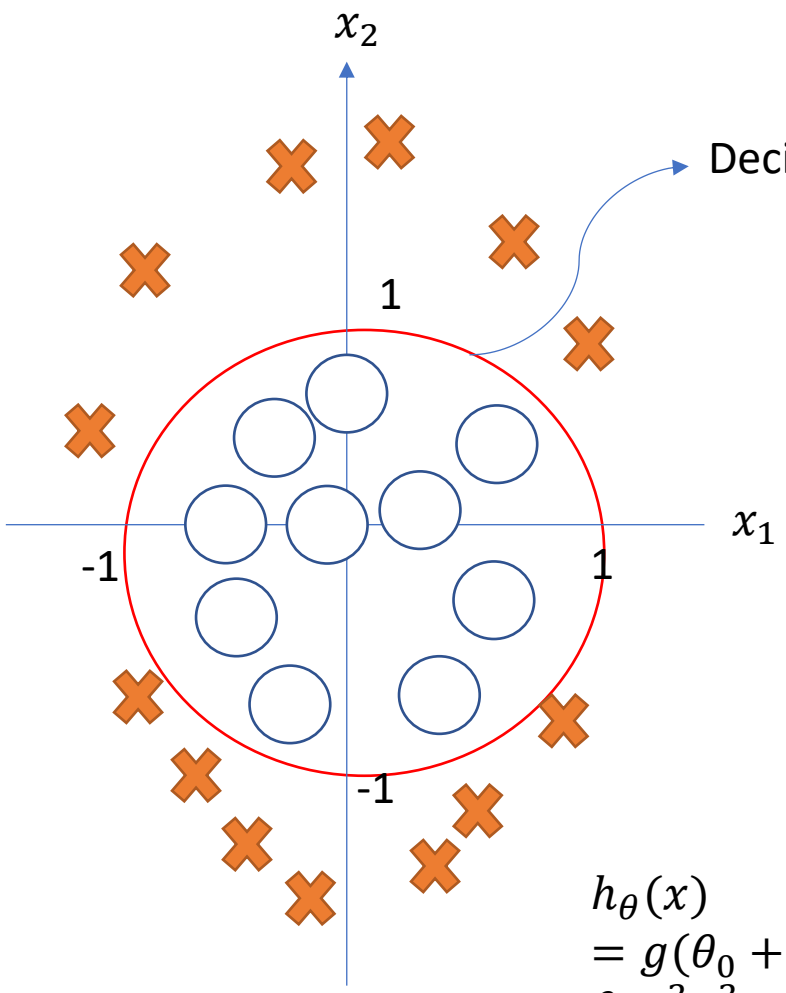
$$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$



Predict $y = 1, if \underbrace{-3 + x_1 + x_2 \geq 0}_{\theta^T X}$

$$x_1 + x_2 \geq 3$$

Non-Linear Decision Boundaries



Decision Boundary

-1 0 0 1 1

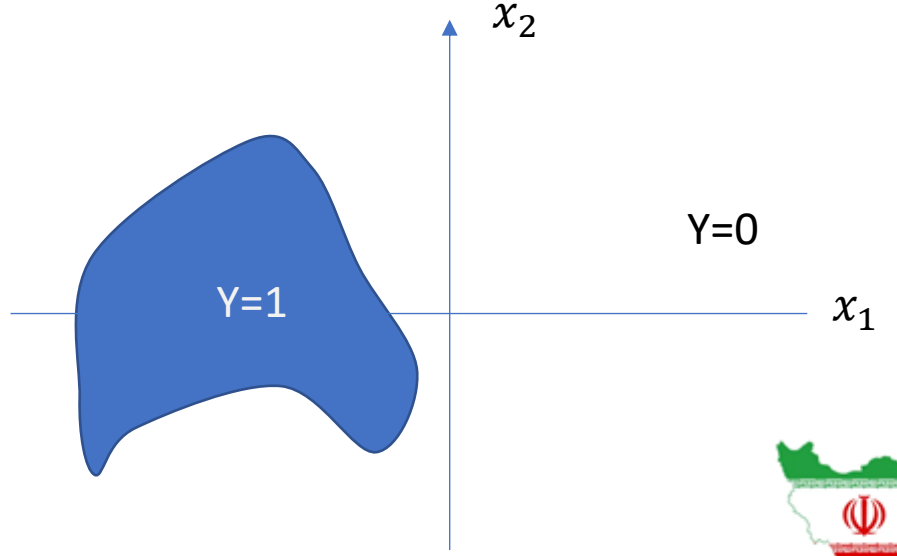
$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

$$\theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Predict $y = 1, if \underbrace{-1 + x_1^2 + x_2^2 \geq 0}_{\theta^T X}$

$$x_1{}^2 + x_2{}^2 \geq 1$$

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \ldots)$$







Cost Function

Training Set: $\{ (x^1 , y^1) , (x^2 , y^2) , \dots , (x^m , y^m) \}$

$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}, \; x_0 = 1, \; y \in \{0,1\}$$

Linear Regression: $J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^i) - y^i)^2$

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

Logistic Regression: $h_\theta(x^i) = g(\theta^T x^i)$

MSE: $J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (g(\theta^T x^i) - y^i)^2$

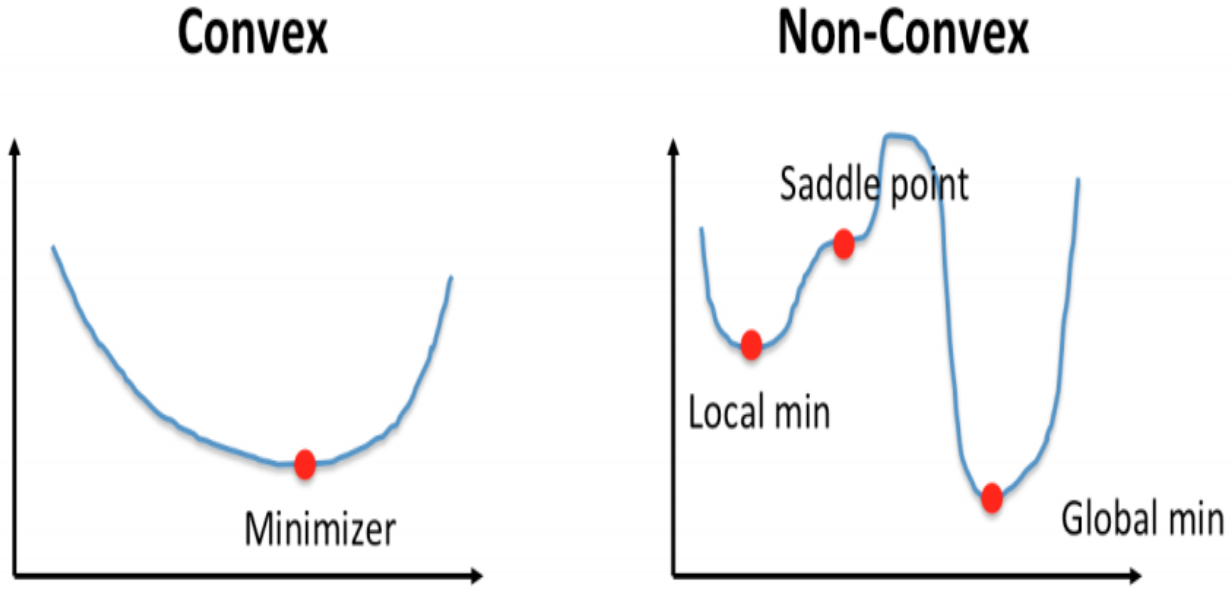




Convex VS Non-Convex Cost Function

Convex Cost Function:
•Bowl-shaped with a single global minimum.
•Easier optimization, guarantees finding the global minimum.

Non-Convex Cost Function:
•Contains multiple local minima.
•**Challenges**: Optimization algorithms can get stuck in local minima.
•**Impact**: Risk of poor model performance, as finding the global minimum is difficult.



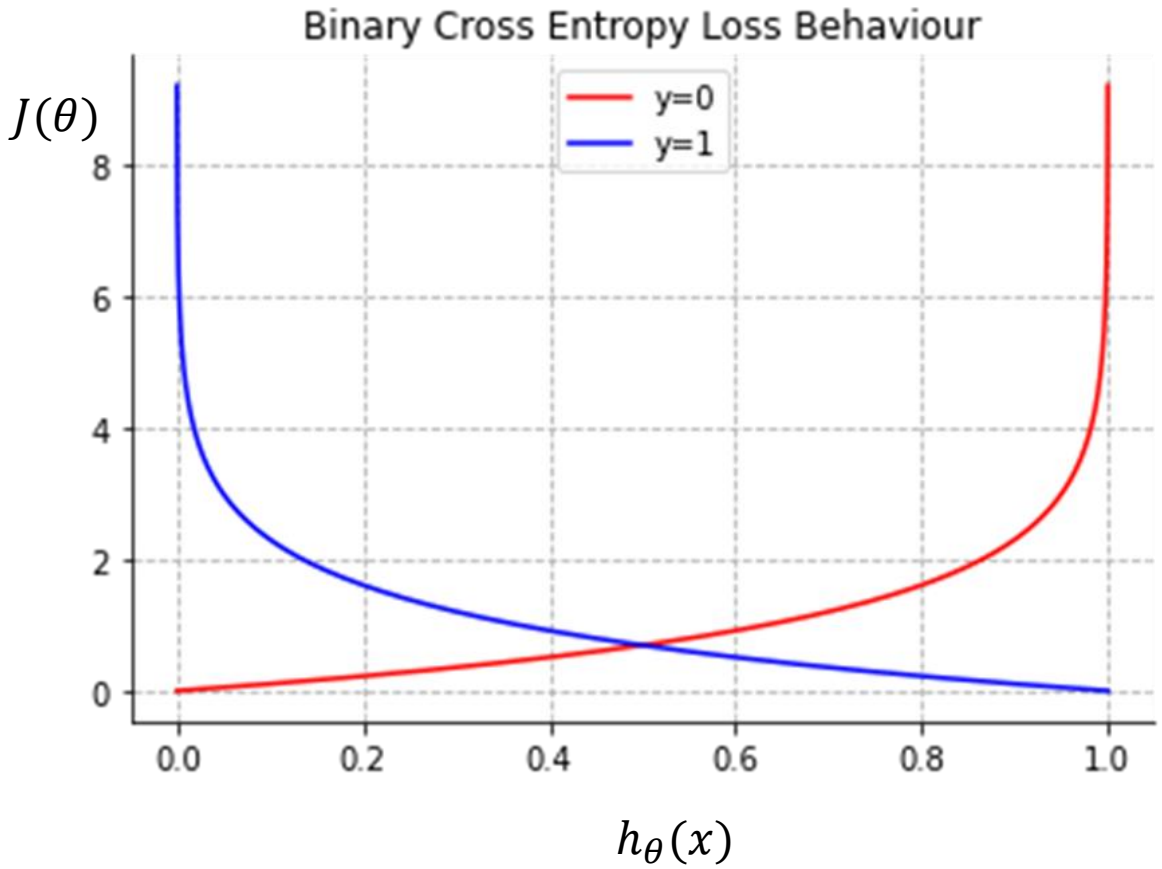
Logistic Regression

MSE Cost Function:

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}((h_\theta(x^{(i)}) - y^{(i)})^2)$$

Binary Cross Entropy cost function:

$$cost(h_\theta(x), y) = \begin{cases} -log(h_\theta(x)), & if \quad y = 1 \\ -log(1 - h_\theta(x)), & if \quad y = 0 \end{cases}$$



if y=1 and $h_\theta(x) = 1 \Rightarrow cost = 0$;

if y=1 and $h_\theta(x) = 0 \Rightarrow cost = \infty$;

if y=0 and $h_\theta(x) = 0 \Rightarrow cost = 0$;

if y=0 and $h_\theta(x) = 1 \Rightarrow cost = \infty$;

Logistic Regression

$$cost(h_\theta(x), y) = \begin{cases} -log(h_\theta(x)), & if \quad y = 1 \\ -log(1 - h_\theta(x)), & if \quad y = 0 \end{cases}$$

$$cost(h_\theta(x), y) = -ylog\big(h_\theta(x)\big) - (1 - y)log(1 - h_\theta(x))$$

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \underbrace{-y^i log\left(h_\theta\left(x^i\right)\right) - (1 - y^i)log(1 - h_\theta(x^i))}_{T_i}$$

$$\min_\theta J(\theta)$$

Gradient Descent: Repeat Unit Convergence :

repeat{

$$\theta_j = \theta_j - \alpha \frac{dJ(\theta)}{d_\theta} \quad J = 0, \ldots, n$$

}until convergence





$$\frac{dJ(\theta)}{d\theta} = ? \qquad J(\theta) = \frac{1}{m}\sum_{i=1}^{m} T_i \qquad \frac{dJ(\theta)}{d\theta} = \frac{1}{m}\sum_{i=1}^{m} \frac{dT_i}{d\theta}$$

$$T_i = -[y^i \log h_\theta(x^i) + (1-y^i)\log(1-h_\theta(x^i)] \qquad h_\theta(x^i) = \sigma(\theta^T x^i) = \sigma(z^i) = \frac{1}{1+e^{-z^i}}$$

$$T_i = -[y^i \log \sigma(z^i) + (1-y^i)\log(1-\sigma(z^i)] \qquad z^i = \theta^T x^i = \theta_0 x_0 + \theta_1 x_1 + \ldots \theta_n x_n$$

$$\frac{dT_i}{d\theta_j} = \frac{dT_i}{d\sigma(z^i)} \times \frac{d\sigma(z^i)}{dz^i} \times \frac{dz^i}{d\theta_j}$$

$$(1)\ \frac{dT_i}{d\sigma(z^i)} = -\left[\frac{y^i}{\sigma(z^i)} + (1-y^i).\frac{-1}{1-\sigma(z^i)}\right] = -\left[\frac{y^i}{\sigma(z^i)} - \frac{1-y^i}{1-\sigma(z^i)}\right]$$

$$(2)\ \frac{d\sigma(z^i)}{dz^i} = \frac{e^{-z^i}}{(1+e^{-z^i})^2} = \frac{1}{1+e^{-z^i}}.\frac{e^{-z^i}}{1+e^{-z^i}} = \sigma(z^i).(1-\sigma(z^i) \quad ,$$

$$1-\sigma(z^i) = 1 - \frac{1}{1+e^{-z^i}} = \frac{1+e^{-z^i}-1}{1+e^{-z^i}} = \frac{e^{-z^i}}{1+e^{-z^i}}$$

$$(3)\ \frac{dz^i}{d\theta_j} = x_j^i$$





From (1), (2) and (3):

$$\frac{dT_i}{d\theta_j} = -\left[\frac{y^i}{\sigma(z^i)} - \frac{1-y^i}{1-\sigma(z^i)}\right]\sigma(z^i).(1-\sigma(z^i))x_j^i$$

$$= -\left[y^i.\left(1-\sigma(z^i)\right) - \left(1-y^i\right).\sigma(z^i)\right]x_j^i$$

$$= -\left[y^i - \sigma(z^i)\right]x_j^i = \left[\sigma(z^i) - y^i\right]x_j^i$$

$$\frac{dJ(\theta)}{d\theta_j} = \frac{1}{m}\sum_{i=1}^{m}\frac{dT_i}{d\theta_j} = \frac{1}{m}\sum_{i=1}^{m}\frac{dT_i}{dz^i}.\frac{dz^i}{d\theta_j} = \frac{1}{m}\sum_{i=1}^{m}\left(\underbrace{\sigma(z^i)}_{h_\theta(x^i)} - y^i\right).x_j^i$$

$$GD: \mathrm{R}peatUntiConvergance\{$$

$$\theta_j = \theta_j - \alpha\frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^i) - y^i\right).x_j^i$$

$$\}$$





Vanishing gradient problem of MSE

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m} T_i \qquad \frac{dJ(\theta)}{d\theta} = \frac{1}{2m}\sum_{i=1}^{m} \frac{dT_i}{d\theta} \qquad T_i = (\sigma(z^i) - y^i)^2$$

$$z^i = \theta^T x^i = \theta_0 x_0 + \theta_1 x_1 + \ldots \theta_n x_n$$

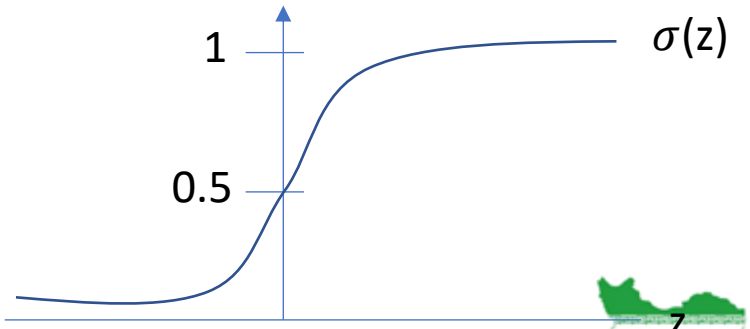
$$\frac{dT_i}{d\theta_j} = \frac{dT_i}{d\sigma(z^i)} \times \frac{d\sigma(z^i)}{dz^i} \times \frac{dz^i}{d\theta_j}$$

$$(1)\ \frac{dT_i}{d\sigma(z^i)} = 2\left[\sigma(z^i) - y^i\right]$$

$$(2)\ \frac{d\sigma(z^i)}{dz^i} = \sigma'(z^i)$$

$$(3)\ \frac{dz^i}{d\theta_j} = x_j^i$$

$$\frac{dT_i}{d\theta_j} = 2\left[\sigma(z^i) - y^i\right]\sigma'(z^i)x_j^i$$



Logistic regression on m examples

$\theta_1 \leftarrow random \quad \theta_2 \leftarrow random \quad b \leftarrow random$

$$\theta^t = \begin{bmatrix} \theta_1^t \\ \theta_2^t \\ b^t \end{bmatrix} \quad \theta^{t+1} = \begin{bmatrix} \theta_1^{t+1} \\ \theta_2^{t+1} \\ b^{t+1} \end{bmatrix}$$

Repeat{

$J = 0; \quad d\theta_1 = 0; \quad d\theta_2 = 0; \quad db = 0;$

$$\|\theta^{t+1} - \theta^t\|_2 \leq \varepsilon$$

For *i=1* *to* *m*

$z^{(i)} = \theta^T x^{(i)} + b$

$a^{(i)} = \sigma(z^{(i)})$

$J \mathrel{+}= [y^{(i)} Log a^{(i)} + (1 - y^{(i)}) Log(1 - a^{(i)})]$

$dz^{(i)} = a^{(i)} - y^{(i)}$

$$d\theta = \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ db \end{bmatrix}$$

$$\|d\theta\| \leq \varepsilon = 10^{-4}$$

$d\theta_1 \mathrel{+}= x_1^{(i)} dz^{(i)} \quad d\theta_2 \mathrel{+}= x_2^{(i)} dz^{(i)} \quad db \mathrel{+}= dz^{(i)}$

$J /= m; \quad d\theta_1 /= m; \quad d\theta_2 /= m; \quad db /= m;$

$\theta_1 = \theta_1 - \alpha\, d\theta_1 \quad \theta_2 = \theta_2 - \alpha\, d\theta_2 \quad b = b - \alpha\, db$

} until convergence

تعبیر احتمالاتی رگرسیون لاجستیک

$$p(y, \mathbf{X}|\theta) = p(\mathbf{X}|\theta)p(y|\mathbf{X}, \theta) = p(\mathbf{X})p(y|\mathbf{X}, \theta)$$

$$L = p(Y|X, \theta) = \prod_{i=1}^{m} p(y_i|x_i)$$

$$= \prod_{n:y_n=1} \underbrace{p(y_i = 1|x_i)}_{\mu_i} \prod_{n:y_i=0} \underbrace{p(y_i = 0|x_i)}_{1-\mu_i}$$

$$= \prod_{i=1}^{m} \left[\underbrace{\sigma(x_i^T\theta)}_{\mu_i}\right]^{y_i} \left[\underbrace{1 - \sigma(x_i^T\theta)}_{1-\mu_i}\right]^{1-y_i}$$

Bernoulli distribution

$$p(y|x) = f(x) = \begin{cases} \mu, & y = 1 \\ 1 - \mu, & y = 0 \end{cases}$$

$$\mu_i = h_\theta(x_i) = \sigma(x_i^T\theta)$$

$$LL = \sum_{i=1}^{m} y_i log\big(h_\theta(x_i)\big) + (1 - y_i)log(1 - h_\theta(x_i))$$

$$j(\theta) = -LL$$

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}}\, j(\theta)$$

برای ساده کردن رابطه و هم چنین مسائل محاسباتی ازتابع log میگیریم. هم چنین یک منفی در آن ضرب می کنیم تا تابع را کمینه کنیم وحالت تابع هزینه پیدا کند.





محاسبات رگرسیون لاجستیک به صورت برداری

$$j(\theta) = -\sum_{i=1}^{m} y_i ln\left(\sigma\left(x_i^T\theta\right)\right) + (1-y_i) ln(1-\sigma\left(x_i^T\theta\right))$$

می توان نشان داد:

$$j(\theta) = \sum_{i=1}^{m} ln\left[1+\exp\left(x_i^T\theta\right)\right] - y_i x_i^T\theta$$

$$\frac{\partial \ln[1+\exp(x)]}{\partial x} = \sigma(x).$$

می توان نشان داد:

$$\frac{d \ln\left[1+\exp(\overbrace{x_i^T\theta}^{z})\right]}{\partial\theta} = d\frac{ln[1+\exp(z)]}{\partial z}\cdot\frac{\partial z}{\partial\theta} = \sigma(z)\frac{\partial x_i^T\theta}{\partial\theta} = \sigma(z)x_i$$

بنابراین:

$$\underbrace{\nabla J(\theta)}_{P\times 1} = \sum_{i=1}^{m} x_i\,\sigma(x_i^T\theta) - x_i y_i$$

$$a^T = y_i x_i^T \qquad a = x_i y_i^T$$

$$\frac{da^T\theta}{d\vec{\theta}} = a$$

محاسبات رگرسیون لاجستیک به صورت برداری

$$\underbrace{\nabla J(\theta)}_{P\times 1} = \sum_{i=1}^{m} x_i \left(\sigma(x_i^T\theta) - y_i\right) = \underbrace{X^T}_{P\times m}\left[\underbrace{\sigma(X\theta)}_{m\times 1} - \underbrace{y}_{m\times 1}\right].$$

$$\alpha_i = \sigma(x_i^T\theta) - y_i \qquad \propto = \begin{bmatrix} \propto_1 \\ \propto_2 \\ . \\ . \\ . \\ . \\ \propto_m \end{bmatrix}$$

$$X = \begin{bmatrix} . & \dots & x^{1\,T} & \dots & . \\ . & \dots & x^{2\,T} & \dots & . \\ . & \dots & x^{3\,T} & & . \\ . & . & . & & . \\ . & . & . & \dots & . \\ . & . & . & & . \end{bmatrix} \in R^{m*(n+1)} \qquad x^i = \begin{bmatrix} x_1^i \\ x_2^i \\ x_3^i \\ . \\ . \\ . \\ x_n^i \end{bmatrix} \in R^{(n+1)*1}$$

$$\alpha_1\overrightarrow{x^1} + \alpha_2\overrightarrow{x^2} + \cdots + \alpha_m\overrightarrow{x^m} = X^T \propto$$





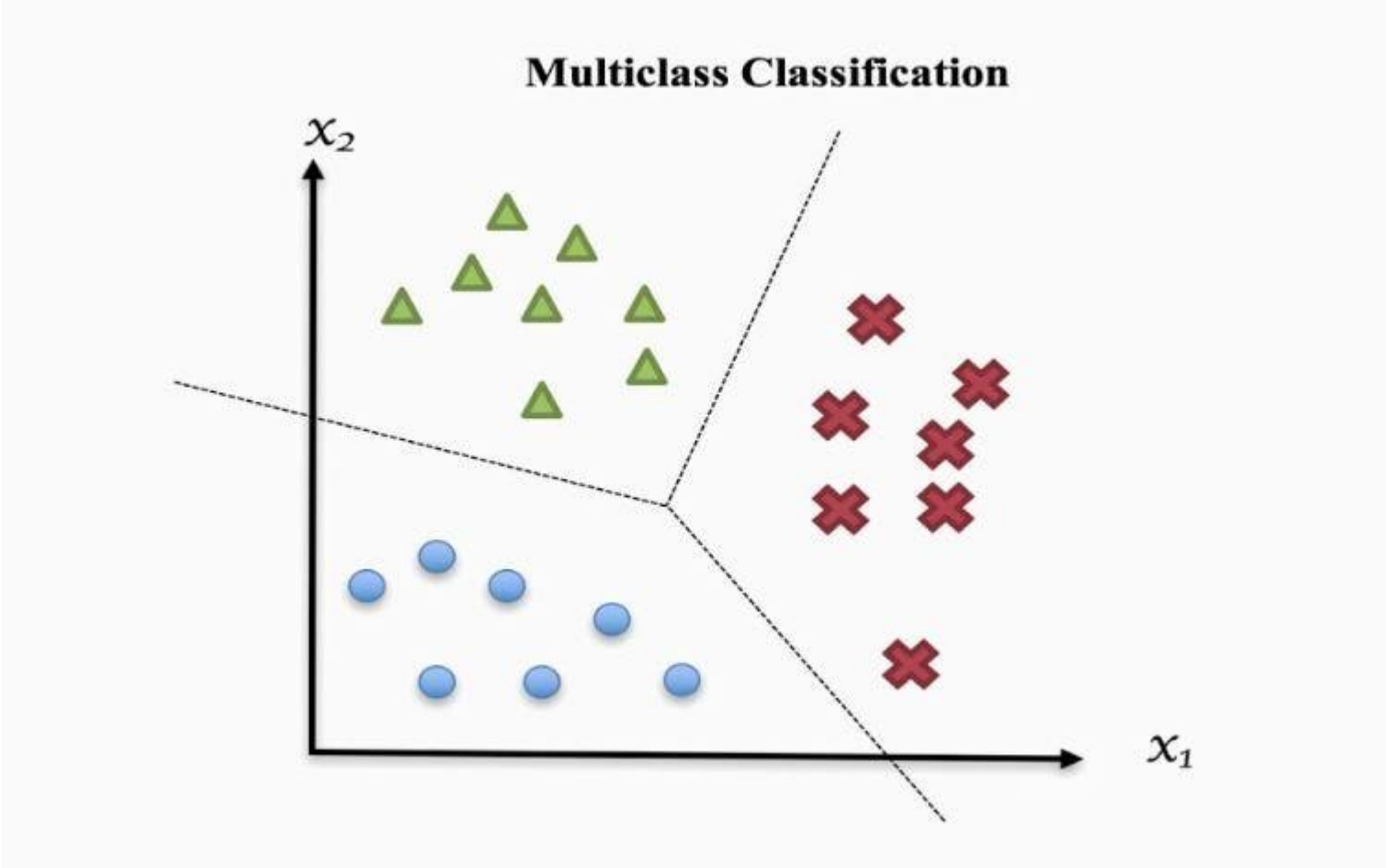
Regularized Logistic Regression

اگرچه حد پایین تابع هزینه رگرسیون لاجیستیک صفر است. ولی در حالتی که داده ها خطی تفکیک نشده باشند هیچ θ محدود این حداقل را به ما نمی دهد. و اگر بهینه سازی را ادامه دهیم θ به بی نهایت میل میکند
(این را نشان دهید) (به عبارتی مسئله ما بیش برازش می شود)
برای اجتناب از این مسئله میتوانیم یک ترم جریمه به تابع هزینه اضافه کنیم :

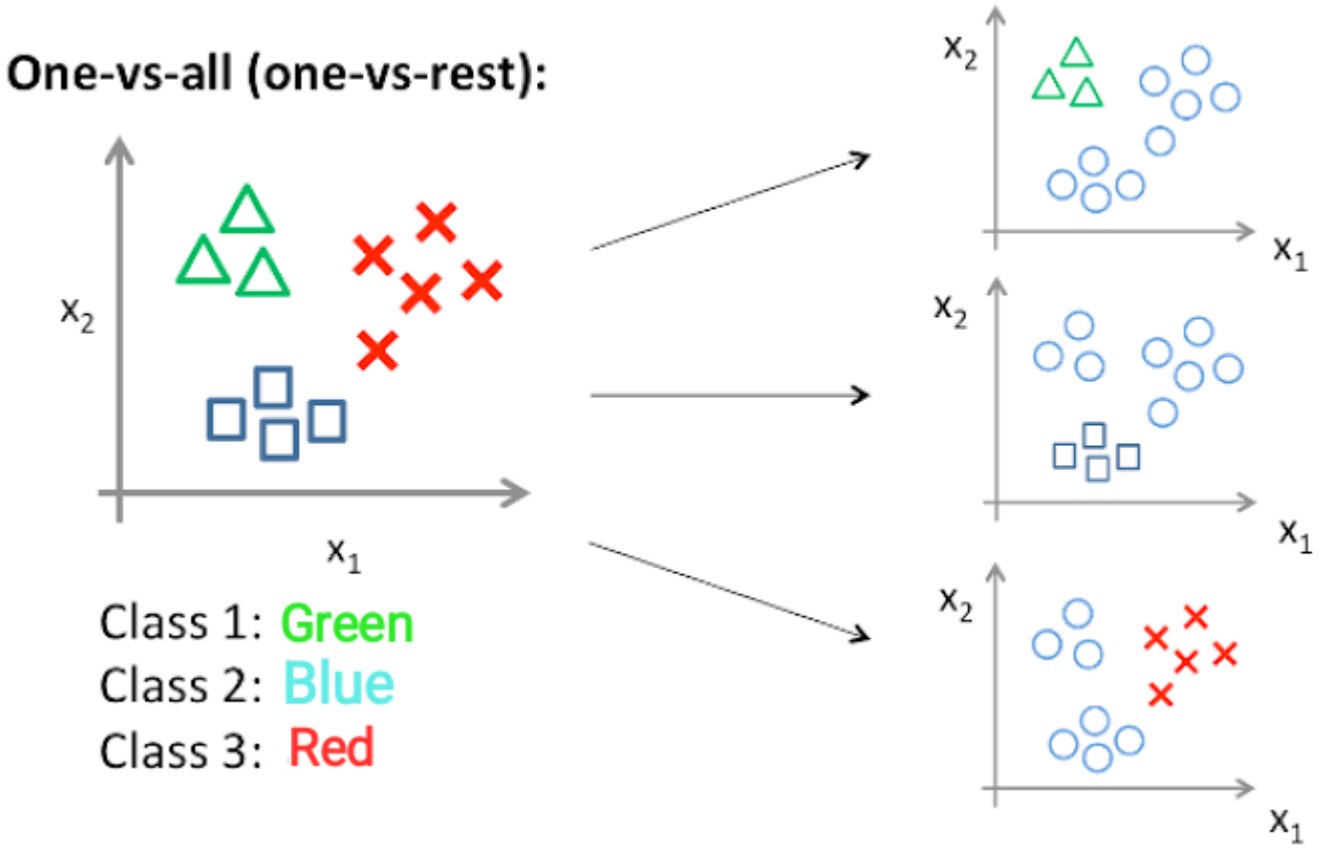
$$argmin_\theta \left(-\sum_{n=1}^{N} \ln p(y_n | x_n^T \theta) + \frac{\boldsymbol{\lambda}}{2} ||\theta||^2 \right)$$

این کار از بی نهایت شدن θ ممانعت می‌کند.

Multi-class Classification



Multiclass Classification: One-Vs-Rest(All)



One-hot-vector

- **Classifier 1:- [Green] vs [Red, Blue]**
- **Classifier 2:- [Blue] vs [Green, Red]**
- **Classifier 3:- [Red] vs [Blue, Green]**

Multiclass Classification: One-Vs-Rest(All)

Main Dataset

Features			Classes
x1	x2	x3	G
x4	x5	x6	B
x7	x8	x9	R
x10	x11	x12	G
x13	x14	x15	B
x16	x17	x18	R

Training Dataset 1
Class :- Green

Features			Green
x1	x2	x3	**+1**
x4	x5	x6	-1
x7	x8	x9	-1
x10	x11	x12	**+1**
x13	x14	x15	-1
x16	x17	x18	-1

Class 1 :- Green Class 2 :- Blue Class 3 :- Red

Training Dataset 2
Class :- Blue

Features			Blue
x1	x2	x3	-1
x4	x5	x6	**+1**
x7	x8	x9	-1
x10	x11	x12	-1
x13	x14	x15	**+1**
x16	x17	x18	-1

Training Dataset 3
Class :- Red

Features			Red
x1	x2	x3	-1
x4	x5	x6	-1
x7	x8	x9	**+1**
x10	x11	x12	-1
x13	x14	x15	-1
x16	x17	x18	**+1**

Test time:

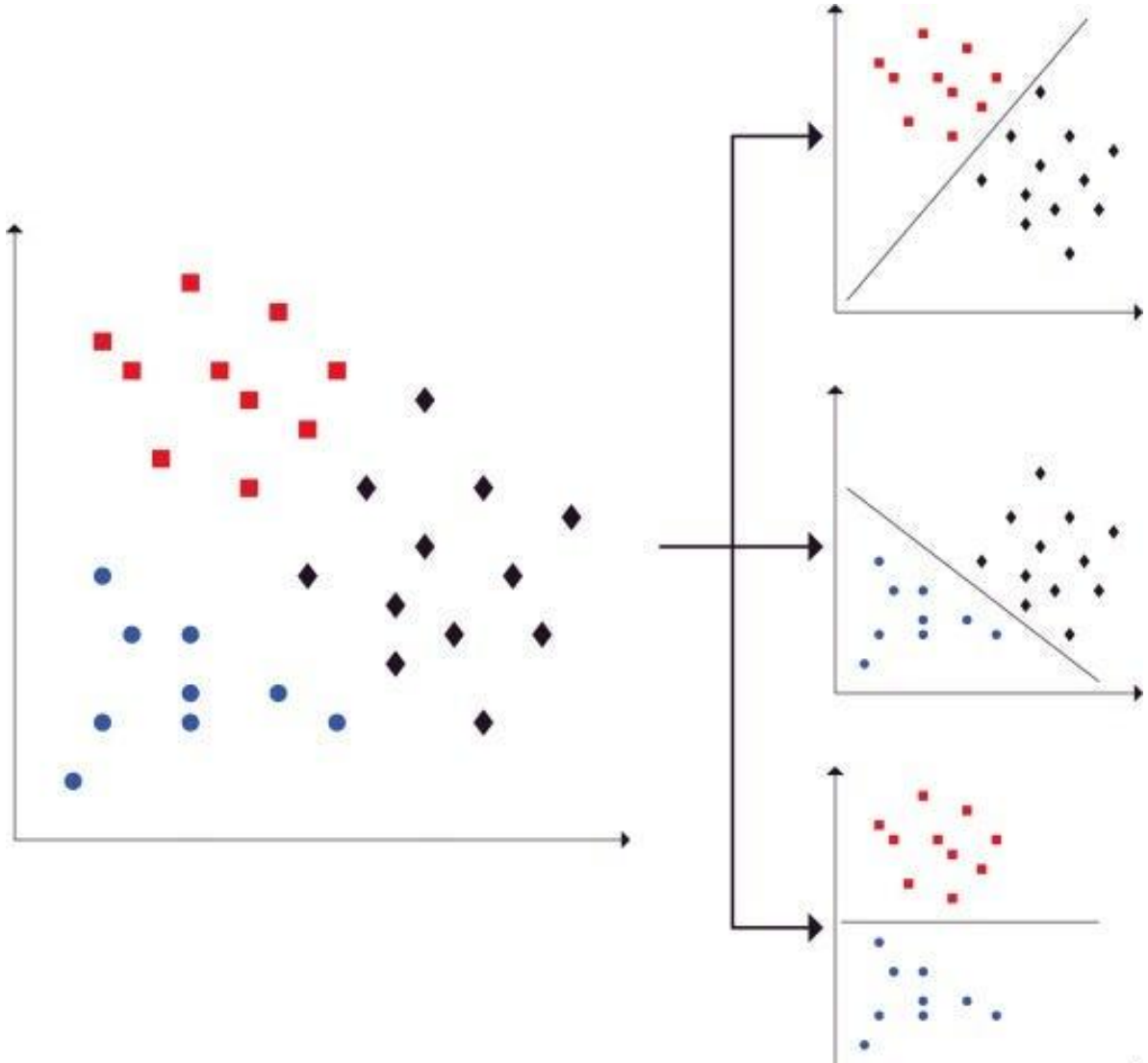
$$h_\theta^i(x) = P(y = i|x)$$

$$\underset{i}{\text{argmax}} h_\theta^i(x)$$





Multiclass Classification: One-Vs-One



N* (N-1)/2 binary classifier models

- Classifier 1: Green vs. Blue
- Classifier 2: Green vs. Red
- Classifier 3: Blue vs. Red

Test time:

Each binary classifier predicts one class label. When we input the test data to the classifier, then the model with the majority counts is concluded as a result.